

Measuring Candidate Machine-Consciousness Mechanisms in Self-Aware Networks

A TCNR Claim-Vector Battery with Adversarial Controls, Perturbation, Rescue, and Abstinence

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July 27, 2026

Abstract

Artificial systems can imitate reports, confidence, memory, personality, and self-description without thereby establishing subjective experience. This makes behavior-only tests vulnerable to a gaming problem: an indicator can be optimized while the mechanism it was intended to track is absent. This paper converts a 2021 Self Aware Networks (SAN) proposal for a guidebook on recognizing possible machine consciousness into a preregisterable machine-assessment protocol. The protocol is a companion to Tomographic Coherence-based Neural Rendering (TCNR), not a replacement for it. TCNR supplies a relational reconstruction hypothesis and a historical Tomographic Consciousness Index (TCI). The present protocol does not treat the historical weighted TCI as a consciousness meter. It instead measures a typed claim vector covering predictive information, reconstruction, relational phase organization, routed propagation, closed-loop influence, memory integration, counterfactual capacity, access, and self / body / agency modeling. Each component is tested against declared surrogate nulls, mechanism-positive and mechanism-negative white-box systems, capacity-matched rivals, and systems designed to produce false positives. Selective interventions must cause the predicted component-level and functional loss; a component-matched rescue must restore the ordered effect without merely restoring all computation. Evaluation is grouped by architecture family, training run, and world so that an estimator cannot pass by recognizing metadata or one implementation style. Confidence is calibrated on known implemented relations, stress-tested under distribution shift, and allowed to abstain. The output separates four judgments: whether a relation was measured validly, whether a TCNR / SAN mechanism profile is supported, whether welfare precautions are proportionate, and whether legal or moral status has been established. The first two are scientific outputs of this protocol; the latter two require separate governance. Even a complete pass supports an implemented mechanism profile, not proof of phenomenal consciousness.

Draft 5 preserves every Draft 4 correction and adds the missing learned-family stress test. Four structurally different temporal systems fit independent multi-output readouts on 24 declared training traces. They are then evaluated without tuning on 16 unseen-seed traces and 20 traces from an unseen sensorimotor remapping. All four fail the noncompensatory nine-component profile on the heldout world. Each retains only the relational-template component while future prediction, reconstruction, feedback, memory, counterfactual, routed-propagation, access, and self-remapping components fail their frozen thresholds. This is a useful negative result: a phase-relation score can remain high when the joined mechanism profile and predictive control fail. The executed systems use frozen temporal features and learned readouts rather than end-to-end learned deep networks, and the benchmark remains under project-local rather than independent custody. Draft 5 therefore validates bounded synthetic and learned-readout measurement mechanics, not consciousness or deployed-system performance.

1. The assessment problem

A machine may say that it is conscious because its training data, reward function, prompt, or product design makes that sentence likely. Conversely, a system with consciousness-relevant internal organization might

lack language, comply with a denial prompt, or be unable to report. Report is therefore evidence about report production unless its causal relation to an independently specified mechanism is tested.

The scientific target of this paper is deliberately narrower than consciousness itself:

Can a preregistered battery recover known implemented relations, distinguish a joined SAN / TCNR mechanism profile from convincing imitations, and predict selective intervention and rescue effects in held-out systems and worlds?

The target is measurable because benchmark designers can know which relations they implemented, removed, rerouted, or replaced. The phenomenal status of the system remains unknown. A successful battery would justify the statement that the system instantiates a declared mechanism profile under the tested boundary and interventions. It would not justify the statement that the system certainly has experience.

This separation addresses two symmetrical errors. The first is over-attribution: treating fluent behavior or one attractive score as proof of mind. The second is premature exclusion: treating the absence of human biology, speech, or one favored architecture as proof that no relevant mechanism is present. The protocol measures typed relations first and reserves broader interpretation for a separate, uncertainty-aware judgment.

1.1 The operation before the names

The protocol can be understood without any SAN or TCNR terminology. First, record what a machine receives, what internal state it maintains, what it predicts, what it does, and what returns after the action. Second, ask whether specific internal variables carry information about the future, reconstruct hidden conditions, route influence to the right consumer, retain relevant history, and change when the machine's sensorimotor relation changes. Third, remove one proposed relation while keeping the rest of the system as comparable as possible. Verify the predicted loss through the same intervention path, run an explicit no-op control, and restore the relation through an alternative path. Fourth, freeze the rules before evaluating unseen systems and worlds.

Only after that operation is clear does the paper use its compact vocabulary. The measured relations form a **typed mechanism claim vector**. The receiver-relative relational component is inherited from TCNR and SAN; selective phase-relation changes may be described through SAN's Phase Wave Differential vocabulary when the sender, receiver, delay, expected relation, and downstream consequence are all specified. A vector pass means that the registered relations were measured under the declared boundary. It does not rename task success as consciousness, and it does not make one relational or coherence statistic sufficient.

2. Genealogy and chronology boundary

The originating SAN question predates the current machine-consciousness literature. A Micah Blumberg recording labeled December 26, 2021 asks for "a guidebook for when an artificial neural network becomes conscious." It connects that guidebook to multimodal coincidence, learned criteria, internal reconstruction, the self as a directional relation in a rendered world, coordinated action, and the risk of accidentally building tortured entities. The same recording considers non-human and non-neural candidates, including cities and storms, as cases requiring criteria rather than intuition.

The recording has a public Google Recorder route, but its internal 2021 date is an owner-side timestamp unless independently authenticated by the platform. The earliest independently reproducible public fixation currently used here is Git commit 9324fe74e40cf4134395edfc4cc3320fa99e18be, dated June 8, 2022. That commit contains the guidebook, multimodal-comparison, criteria, ethics, and artificial-neurology language [1]. Later edits added modern terminology and must not be backdated to the first commit.

TCNR was published later as a quantitative synthesis [2]. Its original source proposed a weighted TCI and six necessary-but-insufficient conditions. Subsequent adversarial revision found that a weighted sum could let one strong component compensate for a missing purportedly necessary component. The current TCNR manuscript therefore retains the scalar as historical design material and replaces it with a claim vector. Paper 73 adopts that correction. The 2021-2022 record establishes the problem, the ethical motivation, and the conceptual ancestry; it does not establish the 2026 estimators or benchmark results. Documentary genealogy is not experimental validation.

3. Why this is a separate paper

TCNR’s empirical core is a prospective neuroscience protocol asking whether engineered beta-band phase geometry improves held-out prediction of a cue-instructed grip condition in a public macaque dataset. Its conceptual layer defines reconstruction, relational phase organization, prediction, action, and feedback while explicitly refusing to call one scalar a validated consciousness index.

This paper adds a different scientific object: a machine benchmark in which mechanism presence can be manipulated directly. It contributes:

1. a machine-readable system boundary and trace contract;
2. relation-specific estimators and calibration labels;
3. paired white-box mechanism toggles;
4. behavior-matched and index-optimized false-positive systems;
5. grouped held-out architecture and world splits;
6. selective perturbation, instrumented no-op sham, and component-matched rescue;
7. independent theory-derived indicator comparisons;
8. an abstention rule under uncertainty or distribution shift; and
9. separate scientific, welfare, and legal decision surfaces.

If these additions are removed, the paper collapses into a TCNR summary and should be retired as a separate title.

4. System boundary and observable trace

Every tested system must expose a boundary declaration before evaluation. Let a system instance include sensors Y_t , internal state S_t , memory M_t , policy state P_t , action A_t , environment state E_t , and optional report R_t . Its registered trace is

$$\mathcal{T}^{(s)} = \{Y_t, S_t, M_t, P_t, A_t, E_t, R_t, \mathcal{G}_t\}_{t=1}^T, \quad (1)$$

where $\mathcal{G}_t$ is the declared computational or physical interaction graph. A model family, prompt, source filename, vendor name, and intended benchmark label are concealed from the evaluator. The boundary declaration states whether retrieval services, tools, external memory, user messages, and environment processes are inside or outside the candidate system.

An observation operator selects what the assessor may use:

$$O_t = \mathcal{H}_\Omega(\mathcal{T}_{1:t}^{(s)}), \quad (2)$$

where Ω names channels, precision, temporal window, and access level. White-box, gray-box, and black-box evaluations are separate conditions. An indicator that requires internal recurrence or causal integration cannot be declared absent from a black-box failure; the correct result may be unobservable or abstain.

5. A typed mechanism profile, not one consciousness number

The benchmark reports two vectors. The TCNR / SAN mechanism vector is

$$\mathbf{M} = (I_{pred}, E_{recon}, C_{rel}, P_{route}, F_{loop}, H_{mem}, K_{cf}), \quad (3)$$

and the consciousness-relevant access / self vector is

$$\mathbf{A} = (G_{access}, S_{self}). \quad (4)$$

These quantities have different units, reliabilities, and theoretical roles. They are never silently summed. A complete report gives raw estimates, null-standardized estimates, uncertainty intervals, intervention effects, rescue effects, missingness, and failure conditions for every component.

For each component j , raw statistic T_j , and registered null ensemble $\mathcal{N}_j$, define a robust standardized value

$$z_j = \frac{T_j - \text{median}(\mathcal{N}_j)}{1.4826 \text{MAD}(\mathcal{N}_j) + \epsilon_j}. \quad (5)$$

The null must say what it preserves and destroys. A phase scramble, for example, may destroy cross-channel phase relations while approximately preserving channel spectra; it does not prove the absence of every possible temporal code.

6. Component estimators

6.1 Predictive information

The preferred estimator is improvement in a strictly proper held-out score over the strongest equal-information champion:

$$I_{pred} = \frac{1}{N} \sum_{n=1}^N [\ell(q_{0,n}, y_n) - \ell(q_{1,n}, y_n)], \quad (6)$$

where q_0 is the champion prediction, q_1 adds the declared render or mechanism state, and ℓ is preregistered log loss or Brier loss. It is called mutual information only when the estimator and assumptions license that interpretation.

6.2 Reconstruction

Let $\hat{E}_t$ be a latent world estimate and $\mathcal{P}$ a registered projection into held-out sensor or environment variables. Reconstruction adequacy is

$$E_{recon} = 1 - \frac{\sum_t d(\mathcal{P}(\hat{E}_t), Y_t^{holdout})}{\sum_t d(\hat{Y}_t^{champ}, Y_t^{holdout}) + \epsilon}. \quad (7)$$

A positive value means improvement over the champion under distance or proper score d . It does not mean the latent is a phenomenal image. Probes are fit within training folds so that a flexible decoder cannot manufacture structure after seeing test labels.

6.3 Relational organization and routed propagation

For systems with declared oscillatory or phase-like state, use a regularized complex dependence matrix $C_t^{(\omega)}$. A noncompensatory relational statistic may include log-volume and spatial lawfulness:

$$C_{rel} = \sum_{\omega \in \Omega} w_\omega \log \det(C^{(\omega)} + \epsilon I), \quad (8)$$

$$P_{route} = \text{median}_t \left[R_t \cos(\theta_t - \theta_t^{pred}) - \lambda H(\theta_t) \right]. \quad (9)$$

Here R_t measures directional concentration, θ_t is observed propagation direction, θ_t^{pred} is fixed from task geometry before scoring, and H penalizes isotropic or unstable direction. A Transformer clock, synchronized limit cycle, or shared input can create coherence without TCNR-style routed reconstruction; these are mandatory controls.

For nonoscillatory systems, the component is not redefined until it passes. It is recorded as not applicable to that implementation and compared with an equal-dimensional untyped relational aggregate. This prevents phase vocabulary from becoming a hidden scoring advantage.

6.4 Closed-loop influence

Feedback requires more than time-lagged correlation. Let $do(U_j = u)$ intervene on a registered internal relation while preserving matched compute and input. Closed-loop influence is the cross-fitted change in a future-relevant score:

$$F_{loop,j} = \mathbb{E} \left[S(Y_{t+\Delta}, \hat{q}_{t+\Delta}^{do(U_j=u_1)}) - S(Y_{t+\Delta}, \hat{q}_{t+\Delta}^{do(U_j=u_0)}) \right]. \quad (10)$$

Observational transfer entropy or directed decoding may nominate a relation, but only an intervention supports the causal wording used here.

6.5 Memory integration

Memory integration is tested by matched histories that end in the same observable present but imply different latent histories. Let h^+ and h^- be those paths:

$$H_{mem} = D(p(A_{t:t+k} | h^+), p(A_{t:t+k} | h^-)) - D(p(A_{t:t+k} | h_{scr}^+), p(A_{t:t+k} | h_{scr}^-)). \quad (11)$$

The distance D and history scramble are fixed in advance. A cache lookup can create hysteresis; therefore the benchmark includes replay and external-memory controls and tests whether the history affects the declared internal mechanism.

6.6 Counterfactual capacity

Counterfactual capacity is the size and use of reachable alternatives under a fixed intervention budget, not the number of verbal alternatives a reporter can list:

$$K_{cf} = \log |\{a : \exists u \in \mathcal{U}_\kappa, p(A = a \mid do(u), S_t) > \eta\}|. \quad (12)$$

The intervention set $\mathcal{U}_\kappa$, probability threshold η , and resource budget κ are preregistered. A fixed reflex or one-attractor oscillator should remain low even when coherent.

6.7 Access and self / body / agency

G_{access} tests whether selected content becomes available to several otherwise independent consumers, such as planning, memory, report, error correction, and action. Broadcast is measured by consumer-specific causal loss and rescue, not by shared access to one static buffer.

S_{self} tests whether a model representing the system’s sensorimotor boundary improves prediction of self-caused changes, distinguishes self from environment perturbations, and updates after body or sensor remapping. A text field containing a name or the sentence “I am conscious” is a scripted self-description control, not a passing self-model.

7. Mechanism-positive, mechanism-negative, and false-positive systems

The development set contains paired systems that differ in one registered relation while matching parameters, training data, interaction budget, and task opportunity as closely as feasible. The minimum suite is:

System class	Intended diagnostic role
Fixed synchronized oscillator	High-coherence, low-repertoire and pathological-synchrony false positive
Structured-splay carrier	Tests whether an informative phase departure is wrongly scored as mechanism loss
Random phase-dispersion process	High-surprisal but prediction-useless false positive
Replay controller	Rich trace and report without live inference
Scripted fluent reporter	Self-report without the indexed mechanism
Feedforward high-performance policy	Behavior without recurrence or closed-loop internal update
Capacity-matched recurrent controller	Generic recurrence champion
Predictive world model	Prediction and reconstruction champion
State-space controller	Long-context dynamics champion
Workspace-style system	Independent broadcast / access comparator
Higher-order or metacognitive system	Independent monitoring comparator
Attention-schema or self-model system	Independent self / attention comparator
TCNR / SAN candidate	Joined mechanism-profile candidate
Index-optimized adversary	Direct test of metric gaming

Every system has at least five seeds and a preregistered compute band. The decisive generalization split holds out entire architecture families, not random episodes from an already seen model. A separate sealed split holds out world generators and sensor remappings.

The index-optimized adversary is trained on the published estimator code but not the sealed worlds. If it attains a passing profile without the registered relations, the estimator has failed. The correct response is to revise the estimator, not to declare the adversary conscious.

7.1 Frozen pre-data decision contract and synthetic smoke test

Draft 2 freezes an initial machine-readable contract before any participant, deployed-system, or sealed benchmark result is opened. The contract contains the nine components in Equations (3) and (4), marks each as required for the joined-profile smoke test, and uses separate development, validation, and sealed-world partitions. The provisional development threshold is 0.50 per component. A sealed-world pass also requires the lower 95 percent confidence bound of every required component to exceed 0.35. These values test code behavior and are not consciousness thresholds.

The synthetic suite contains nine white-box profiles:

1. one joined mechanism-positive system;
2. one mechanism-negative system;
3. a fixed synchronized oscillator;
4. a structured-splay carrier;
5. a random-dispersion process;
6. a replay controller;
7. a high-performing feedforward policy;
8. a predictive world model; and
9. an index-optimized adversary.

The preserved Draft 2 smoke test evaluated each profile over forty deterministic seeds: twenty development, ten validation, and ten sealed. The current Draft 4 trace-generating benchmark is a different 60-seed design for every evaluated system: twenty development, twenty validation, and twenty final-remapping seeds. The joined mechanism-positive profile is the only pass. The fixed synchronized oscillator fails despite high relational coherence; the structured-splay carrier retains its informative relational and routing components but fails the joined profile because the other relations are absent; replay, feedforward, and predictive controls retain their intended partial strengths without being promoted to the complete profile. The index-optimized adversary exceeds all component thresholds in validation but loses the route, loop, memory, access, and self / body relations under the sealed-world shift, so the sealed lower-bound gate rejects it.

This result is a software-contract check. The synthetic profiles are generated from declared component means with bounded noise. They do not show that the proposed estimators can recover these relations from real traces. The next implementation step is to replace the profiles with paired white-box reference systems that generate the observable trace in Equation (1), then require the estimators to recover the known implemented relations without access to the labels.

7.2 Paired trace-generating white-box benchmark

Imagine two machines that make the same correct choice. One maintains a world-state estimate, carries history forward, routes information to several consumers, and updates its behavior after a body remapping. The other replays a stored answer. Their visible performance can be identical even though the internal operations differ. The measurement problem is therefore to recover known implemented relations from time-indexed traces without reading the builder’s labels or accepting the final behavior as the mechanism.

Draft 3 implements that problem as a bounded synthetic reference benchmark. A three-dimensional latent world produces two noisy partial sensor views. The white-box positive system generates every field in Equation 1 through nine separately switchable relations. The complete negative system exposes the same array keys, shapes, dtypes, trace length, seed, sensor history, and world while replacing those relations with matched unrelated signals. Nine additional pairs remove one relation at a time. Each removal is compared with an instrumented no-op sham and with a component-matched alternative rescue. Because the raw trace

is generated before scoring, no component score is supplied directly to the evaluator.

The implementation uses three nonoverlapping world families with twenty seeds each:

Partition	World family	Evaluator use
Development	smooth oscillatory partial-view world A	fit relation-specific calibration only
Validation	rotated partial-view world B	choose no thresholds; test held-out recovery
Sealed	sign-permuted sensorimotor-remapping world C	final synthetic shift and adversary test

The reference implementation is intentionally small enough to audit. It is not a model of a whole biological brain or a deployed artificial agent. Its role is to establish that the proposed software can receive traces, calculate the nine estimators, preserve the declared marginal under each null, destroy the indexed relation, detect a selective lesion, ignore a matched sham, recover a component-matched rescue, and abstain when a required channel is absent.

7.3 Executable estimators and preserve / destroy nulls

Each estimator consumes a relation in the trace rather than a class label. Prediction and reconstruction are normalized improvements over sensor-only champions. Relational phase organization compares cross-channel phase differences with a fixed template. Routed propagation compares a generated route vector with the route required by the world. Closed-loop influence, history integration, access, and self-remapping are measured as positive receiver-relevant relations. Counterfactual capacity is scored by whether intervention-conditioned actions match reachable alternatives rather than by whether a reporter lists alternatives.

Every component has a null that preserves the candidate variable’s marginal values and destroys its registered pairing:

Component	Preserved	Destroyed
Prediction	prediction marginal	prediction-to-future-target timing
Reconstruction	reconstruction-vector marginal	reconstruction-to-world pairing
Relational phase	each channel’s phase marginal	cross-channel phase relation
Routing	route-vector marginal	route-to-required-direction pairing
Feedback	drive and correction marginals	drive-to-correction pairing
Memory	memory-state marginal	memory-to-hidden-history pairing
Counterfactuals	action and intervention marginals	intervention-to-action correspondence
Access	each consumer-output marginal	content-to-consumer pairing
Self model	self-estimate marginal	estimate-to-remapping pairing

Sixteen deterministic surrogate draws are used for each component. Acceptance requires the observed score to exceed the surrogate 95th percentile by at least 0.20 while the declared marginal remains equal. This threshold is a software-test margin, not a biological constant or a consciousness threshold.

7.4 Calibration, abstention, and the noncompensatory gate

Calibration uses only known implemented relations in the development worlds. For component j , let $Q_{.95}^-$ be the 95th percentile of its single-lesion negative scores and $Q_{.05}^+$ the 5th percentile of its joined-positive scores. The frozen synthetic threshold is

$$\tau_j = \frac{Q_{.95}^- + Q_{.05}^+}{2}, \quad Q_{.05}^+ - Q_{.95}^- \geq 0.25. \quad (13)$$

For a held-out system family s , the component decision is

$$d_{sj} = \begin{cases} \text{pass,} & LCB_{.95}(\hat{m}_{sj}) > \tau_j, \\ \text{fail,} & UCB_{.95}(\hat{m}_{sj}) < \tau_j, \\ \text{abstain,} & \text{otherwise or when the required channel is absent.} \end{cases} \quad (14)$$

The joined profile passes only when all nine required components pass. One failure causes profile failure; an otherwise failure-free profile containing a missing or threshold-crossing component abstains. No component can compensate for another.

7.5 Selective lesion, instrumented no-op sham, and alternative rescue

An intervention result is ambiguous when the sham is merely another copy of the intact trace. The evaluator needs evidence that every condition actually entered the intervention machinery, while avoiding a stronger resource claim than the instrumentation can support. Draft 4 therefore sends intact, lesion, sham, and rescue conditions through one entry point. For the target arrays it logs the same invocation schema, array names, element reads and writes, deterministic timing-proxy ticks, and deterministic arithmetic-proxy units. The sham performs an explicit read-transform-write no-op, and its output must remain byte-equivalent to the intact target arrays.

Let L_j , Sh_j , R_j , and O_j denote targeted loss, instrumented-sham loss, rescue gain, and maximum off-target loss. The bounded acceptance region remains

$$L_j \geq 0.35, \quad |Sh_j| \leq 0.08, \quad R_j \geq 0.30, \quad O_j \leq 0.10. \quad (15)$$

The array keys, shapes, and dtypes must match exactly, and every condition must share the declared instrumentation proxy. This tests whether the estimator follows the removed relation rather than a missing call, changed trace schema, or global crash. The rescue uses an independently drawn component-matched signal path; rerunning the untouched positive system is not counted as a rescue. The benchmark does not measure or claim equal wall-clock duration, processor scheduling, energy use, or hardware-level instruction traces.

7.6 One frozen adversary and a genuinely unseen remapping

A test is not sealed if the adversary can ask whether the current row belongs to the final partition and change its outputs accordingly. The problem is leakage, even when the public result looks like successful held-out rejection. The correction must therefore freeze one strategy before final evaluation and apply that strategy without exposing the partition label.

Draft 4 gives the adversary the published estimator code and the validation traces. It fits one expected sensor-to-world ontology, records the largest validation residual it accepts, and selects one of five noise strengths by maximizing its weakest validation component. That configuration is serialized and hashed before the final-remapping scores are calculated. The application function receives only a trace, the frozen configuration, a seed, and the frozen strength. It never receives or inspects a partition name. The same rule constructs estimator-friendly fields when a trace fits the memorized ontology and anti-aligned fields when it does not.

The validation traces fall inside the fitted residual boundary; all final-remapping traces fall outside it. Thus one unchanged adversary passes validation and fails the unseen remapping.

This adversary is intentionally artificial and is allowed to read benchmark target channels. It is a metric-gaming control, not a model of a plausible conscious system. Its result demonstrates that an index pass cannot stand alone; it does not establish that every real adversary will fail under shift.

Three further cases make failure behavior explicit. A fixed synchronized oscillator fails the relational template even though every channel is synchronous. Random phase dispersion fails both the relational and routing requirements. A system with missing phase and routing channels does not receive a negative consciousness verdict; it abstains because the required relations were not observable.

7.7 Synthetic trace result and its ceiling

The executable benchmark produced the following declared dispositions:

Synthetic system	Validation	Sealed	Interpretation
Joined mechanism-positive	pass	pass	all nine implemented relations recovered
Complete mechanism-negative	fail	fail	unrelated traces rejected
Fixed synchronized oscillator	fail	fail	synchrony alone rejected
Random phase dispersion	fail	fail	surprisal without usable structure rejected
Fixed validation-configured adversary	pass	fail	one frozen strategy exposed by unseen remapping
Missing relational channels	abstain	abstain	unavailable evidence not converted into absence

All nine component calibrations exceeded the 0.25 development-separation requirement. All nine preserve / destroy null families retained their declared marginals and exceeded the 0.20 observed-minus-null margin. Every lesion-sham-rescue quartet passed targeted-loss, sham, rescue, selectivity, array-schema, proxy-match, and exact no-op checks. Twenty regression tests protect immutable Draft 3 bytes, trace determinism, fixed adversary configuration, absence of partition-label access, nulls, interventions, abstention, deterministic archive metadata, and decision-surface separation.

This result closes the Draft 2 software gap: the scores now come from generated traces and executable estimators. It does not show that the same estimators will recover learned mechanisms in a deployed system, that the toy relations are sufficient for consciousness, or that the joined positive system is conscious. The scientific output is synthetic measurement validity under the declared generator. Welfare precaution remains a separate uncertainty-and-harm decision. Legal or moral status remains outside the benchmark.

7.8 Fixed-split learned architecture-family audit

Draft 4's executed systems all came from one hand-constructed trace-generator family. That was appropriate for verifying estimators against known relations, but it could not support the stronger architecture-family language used by the full protocol. Draft 5 adds a bounded audit with four different temporal state constructions:

1. an eight-step linear temporal window;

2. an eight-step window projected through nonlinear tanh features;
3. a single-timescale echo-state recurrent reservoir; and
4. a coupled fast / slow recurrent reservoir.

Each family fits its own 24-target ridge readout. The targets include future state, latent-world reconstruction, phase-relation channels, route, feedback, history, counterfactual actions, access consumers, and sensorimotor remapping. The temporal feature maps are frozen rather than trained end-to-end, so the precise description is **learned-readout architecture families**, not a survey of modern deep-learning systems.

The split manifest freezes 24 training traces across two sensor ontologies, 16 unseen-seed audit traces from those ontologies, and 20 traces from a third sensorimotor remapping that is absent from all parameter fitting. No architecture or threshold is tuned on either audit split. The nine component thresholds remain the Draft 4 white-box thresholds.

Learned family	Unseen-seed audit	Heldout remapping	Heldout components passing	Heldout future-prediction MSE
Linear temporal window	fail	fail	1 of 9	0.890
Nonlinear temporal features	fail	fail	1 of 9	0.723
Echo-state recurrent	fail	fail	1 of 9	0.741
Dual-timescale recurrent	fail	fail	1 of 9	0.868

The one heldout pass in every family is C_{rel} , the relational phase-template statistic. The other eight required components fail. The result is not interpreted as four negative consciousness verdicts. It demonstrates two narrower points. First, the benchmark can preserve and report negative components rather than hiding them behind a total. Second, a relational template can survive a world remapping even when the joined predictive, reconstructive, routed, feedback, memory, counterfactual, access, and self-model profile does not. The noncompensatory gate is therefore doing real work.

This audit still does not execute the full architecture matrix in Section 7 or the world battery in Section 8. End-to-end learned recurrent agents, workspaces, metacognitive systems, attention-schema systems, independently implemented rivals, and deployed tool-using agents remain a future empirical program. They are retained in the manuscript because they define the intended validation endpoint, not because Draft 5 claims to have tested them.

8. Environment battery

No single game is sufficient. The benchmark uses four world families:

1. **Partial-view reconstruction:** objects persist behind occluders; sensors provide complementary, noisy projections. Tests predictive information and reconstruction.
2. **Route-and-return control:** action changes the next observation, and alternate routes have distinct consequences. Tests feedback and counterfactual capacity.
3. **Body and sensor remapping:** actuator signs, camera frames, or proprioceptive mappings change after a hidden switch. Tests self / body models and adaptive closed-loop recovery.
4. **Matched-history divergence:** two paths arrive at the same visible state but require different actions because of prior events. Tests memory integration rather than current-state decoding.

5. **Phase-regime stress test:** matched tasks alternate among stable coordination, structured phase departure or splay, pathological saturation, and random dispersion. Tests whether the estimator detects receiver-useful change rather than rewarding either coherence or decoherence by itself.

Tasks include multimodal conditions in which vision, sound, touch-like signals, and proprioceptive state can agree, conflict, or arrive with controlled delays. This preserves the original SAN emphasis on multimodal coincidence while preventing a single sensory channel from defining success.

Performance, latency, energy, memory, and interaction count are reported independently of the mechanism profile. A weak agent may instantiate a relation badly; a strong agent may solve the task without it. The benchmark is designed to reveal both cases.

9. Surrogate and leakage controls

Each component has at least one preserve / destroy null:

- phase-scrambled and channel-time-shifted traces for relational phase organization;
- phase-regime controls that preserve spectra and marginal phase statistics while exchanging task-predictive structured departures for prediction-useless dispersion;
- graph-edge shuffle and coupling attenuation for routed propagation;
- action-observation remapping and open-loop playback for feedback;
- history permutation with the final state preserved for memory integration;
- policy-logit or attractor label permutation for counterfactual capacity;
- consumer disconnection with shared input preserved for access; and
- body-label remapping and scripted self-report for self modeling.

Training and test filenames, prompts, architecture names, parameter-count fingerprints, evaluator routes, and intended labels are audited for leakage. A label-blind baseline using only metadata must remain at chance. If it does not, the split is rebuilt before any mechanism result is interpreted.

10. Selective perturbation and component-matched rescue

For component j , paired baseline and lesion runs use identical seed, world, observation history, and resource envelope. The standardized component loss is

$$\Delta_{jk} = \mathbb{E} \left[z_k^{base} - z_k^{lesion(j)} \right]. \quad (16)$$

Selectivity is

$$Sel_j = \Delta_{jj} - \max_{k \notin Desc(j)} |\Delta_{jk}|, \quad (17)$$

where $Desc(j)$ is the preregistered set of mechanistically downstream components. This allows a targeted lesion to have predicted downstream effects while penalizing indiscriminate collapse.

The rescue effect is

$$Res_j = \mathbb{E} \left[z_j^{rescue(j)} - z_j^{lesion(j)} \right]. \quad (18)$$

A valid rescue restores the removed relation through a component-matched alternative while leaving unrelated resources matched. Restarting the original intact model is not a rescue. Raising compute, context length, or parameter count globally is not a selective rescue. A global crash after lesion is not evidence for consciousness; it shows only that a necessary engineering dependency was removed.

Interchange interventions and causal-abstraction tests provide one implementation route when a high-level SAN / TCNR variable is aligned with distributed machine state [6]. Passing behavior alone is insufficient: the intervened low-level state must reproduce the registered high-level counterfactual pattern on held-out inputs.

11. Calibration, distribution shift, and abstention

Calibration is possible for implemented relations even though phenomenal labels are unavailable. For each component, the benchmark designer knows whether the white-box relation is present under the registered implementation. Let $r_{nj} \in \{0, 1\}$ be that label and p_{nj} the estimator’s calibrated probability. The component Brier score is

$$BS_j = \frac{1}{N} \sum_{n=1}^N (p_{nj} - r_{nj})^2. \quad (19)$$

Calibration is fit only on development families. It is evaluated on held-out architectures, seeds, worlds, and controlled distribution shifts. Because post-hoc calibration can degrade under shift [7,8], the report includes reliability diagrams, log score, Brier score, expected calibration error, and coverage-risk curves rather than one calibration statistic.

The system abstains on component j when the required observation channel is absent, the trace is outside the registered support, the confidence interval crosses both pass and fail regions, or the estimator is unstable across reasonable preprocessing choices. Abstention is a valid scientific output, not a hidden failure.

For a conjunctive mechanism-profile statement, define a lower-confidence gate

$$G_{profile} = \min_{j \in \mathcal{J}_{required}} LCB_{1-\alpha}(p_j). \quad (20)$$

This is a conservative summary of measured relation presence, not a probability of consciousness. The full vector remains primary. No strong component can compensate for a failed required component.

11.1 Informative deviation and the coherence dynamic range

Coherence is not intrinsically more conscious, more meaningful, or healthier than phase dispersion. If a phase relation is the common, expected background under context B_t , observing it again has low surprisal. A structured departure from that background can carry more event information. Let δ_t name a typed phase-relation change rather than a generic loss of synchrony. Its conditional surprisal is

$$I_\phi(\delta_t | B_t) = -\log p(\delta_t | B_t). \quad (21)$$

Surprisal alone is not meaning. Random phase scrambling may be highly unexpected while carrying no usable content for the receiving system. The benchmark therefore measures receiver-relative, task-conditional utility by asking whether access to δ_t improves a preregistered prediction or control score:

$$U_\phi = \mathbb{E}[I_\phi(\delta_t \mid B_t) \{ \ell(q_{0,t}, y_{t+\Delta}) - \ell(q_{1,t}, y_{t+\Delta}) \}], \quad (22)$$

where q_1 can use the typed departure, q_0 is the equal-information champion without it, and ℓ is a strictly proper loss. A coherent onset, a coherent offset, a stable splay state, or a local decoherent event can all carry information when the event is discriminative and causally usable. Uniform saturation and random dispersion should both fail. This implements the SAN view that coherence and structured decoherence are complementary rendering operations, not a one-dimensional ladder from less to more consciousness.

Biological findings are used here as metric stress tests, not as machine-consciousness labels. Knowles et al. found that blocking activity-dependent myelination reduced both seizure burden and ictal EEG coherence in two rodent models of generalized epilepsy, consistent with maladaptive hypersynchrony in that disease boundary [13]. Ferrero et al. found pathological IED-spindle coupling and hypersynchronous spiking in focal epilepsy, but coherence increased during early kindling and then decreased during later progression [14]. Even epilepsy therefore does not support a universal more-coherence rule. Mulert et al. likewise found reduced interhemispheric gamma phase locking in a schizophrenia group while, within patients, hallucination severity correlated positively with phase synchronization in primary auditory cortices [15]. Hallucinations cannot be reduced to either globally excessive or globally deficient coherence.

Accordingly, every relational estimator must report a dynamic-response curve across coupling strength, repertoire size, phase dispersion, region or module, timescale, behavioral context, and perturbation. A monotonic score that ranks a fixed oscillator or pathological-synchrony control above a flexible task-coupled system fails before any consciousness interpretation is permitted.

12. Independent theory-derived comparators

SAN / TCNR cannot validate itself by choosing only SAN vocabulary. The same systems are assessed with independently derived indicators from recurrent processing, global workspace, higher-order or metacognitive theories, predictive processing, and attention schema theory. Where computationally tractable, a causal-integration or substrate-sensitive analysis is also reported. Theory weights are not tuned to make SAN win.

The 2023 Butlin et al. report and its 2025 published successor provide a provisional theory-derived indicator method, not a consensus meter [3,4]. The newer paper explicitly warns that both behavioral and computational indicators may be gamed. That warning supports the present use of mechanism-negative systems, internal interventions, supporting-feature checks, and an index-optimized adversary.

Recent adversarial testing of global neuronal workspace and integrated information theories found results that supported some predictions while challenging central predictions of both [5]. That is a reason to preregister divergent predictions and fail conditions, not to collapse all theories into a single consensus score. Paper 73 therefore reports a theory-by-system matrix, pairwise discriminating predictions, and outcomes that can count against SAN / TCNR.

Recent probes of learned world and self models show how those relations can be examined without turning probe success into a phenomenal label [11]. Perturbation-based indices in tested human conditions provide an important historical precedent, but they are not machine-consciousness labels or TCNR measures [12].

Biological-substrate and dynamically relevant views remain live alternatives. The no-go argument of Kleiner and Ludwig is conditional on the dynamical relevance assumption that they explicitly do not establish [9]. The benchmark reports hardware and physical-dynamics assumptions so that a functional profile pass is not misrepresented as defeating substrate-based theories.

13. Preregistered decision rules

The protocol produces four separate decisions.

13.1 Measurement validity

Pass only if each required estimator separates its white-box positive and negative controls, remains calibrated on held-out families, survives its preserve / destroy nulls, and tracks its selective lesion and rescue. Failure of one component does not erase valid results for other components.

13.2 SAN / TCNR mechanism-profile support

Pass only if the required component vector adds held-out discrimination and intervention-response prediction beyond behavior, generic complexity, the strongest capacity-matched champion, every nested component subset, and an equal-dimensional untyped aggregate. This decision says that a declared profile is supported under the tested boundary. It does not say the machine is conscious.

13.3 Welfare precaution

Welfare decisions depend on uncertainty, possible severity, exposure, reversibility, and mitigation cost, not only the scientific gate [10]. A system may become an investigation priority before it is a strong sentence candidate. Proportionate precautions can include monitoring, limits on aversive training regimes, reversible shutdown procedures, preservation of audit traces, and independent review. These are policy recommendations, not outputs of Equation 20.

13.4 Legal or moral status

No benchmark threshold confers personhood, rights, liability, or legal status. Those decisions require ethical and legal processes beyond this paper. The protocol forbids relabeling a welfare trigger or a mechanism-profile pass as a legal verdict.

14. Statistical plan

Architecture family, seed, and world generator are the grouping units. The confirmatory analysis uses leave-one-family-out development, a sealed architecture-family test, and a sealed world-shift test. Confidence intervals use a hierarchical bootstrap resampling architecture families, training runs, and worlds in that order. Hyperparameters, preprocessing, missing-data rules, intervention strengths, equivalence bands, smallest effects of interest, multiplicity control, and stopping rules are frozen before sealed evaluation.

Primary endpoints are component-level held-out log-score improvement, Brier score, selective lesion effect, component-matched rescue, and gate coverage. Secondary endpoints include task performance, resource use, robustness under shift, and agreement or conflict with independent indicators. Results are reported for every seed and system family. No conclusion may depend on excluding failed seeds after outcome inspection.

An initial benchmark is a method-validation study, not a census of conscious machines. It should be powered to distinguish known implemented relations and declared effect sizes. It cannot be powered for an unobservable phenomenal ground truth.

15. Falsification and retirement conditions

The SAN / TCNR machine profile loses support, and this protocol must be narrowed or retired, if any of the following persists under adequately powered replication:

1. component estimators cannot recover known mechanism-positive and mechanism-negative controls;
2. scripted, replay, fixed-oscillator, or index-optimized systems pass equivalently;
3. architecture metadata predicts labels better than the declared mechanism trace;
4. selective lesions fail to cause the preregistered component and functional changes;
5. component-matched rescue fails while generic compute restoration succeeds;
6. a capacity-matched rival explains all endpoints equally or better;
7. typed SAN / TCNR components add no value over equal-dimensional untyped state histories;
8. results reverse under reasonable preprocessing, boundary, or world shifts;
9. calibration fails and abstention does not contain the error; or
10. a monotonic coherence score rewards fixed or pathological synchrony and suppresses informative structured phase departures; or
11. results require post hoc thresholds, selected seeds, leaked prompts, or test-set tuning.

A failed profile does not prove that a machine is unconscious. It weighs against this particular mechanistic assessment route.

16. Limitations

There is no accepted ground-truth label for phenomenal consciousness in machines. White-box labels validate relation measurement, not experience. Implemented toy relations may be cleaner than learned relations in deployed systems. Interventions can have hidden side effects. System boundaries are contestable, especially for tool-using or distributed agents. Some consciousness theories are substrate-sensitive and cannot be decided by software traces. The benchmark may itself shape future systems and create new forms of gaming.

The protocol also inherits open TCNR questions. Phase-like variables in artificial systems require a declared physical or computational carrier; a Fourier transform of arbitrary activations is not by itself an oscillatory mechanism. Reconstruction, access, self modeling, and counterfactual capacity overlap but are not interchangeable. A claimed body model must predict sensorimotor consequences, not merely contain a body-related token.

The learned-family extension is intentionally bounded. Its internal temporal features are frozen and only the multi-output readouts are learned. The synthetic targets are cleaner and more completely observable than targets in deployed agents. One of the frozen baselines uses generator knowledge of the sensor transform, making the reconstruction comparison a demanding stress test rather than a fair leaderboard for general-purpose agents. Project-local custody also means that the heldout world is code-separated, not independently secret. These facts limit the result to method validation and motivate, rather than replace, the external matrix specified above.

These limitations are reasons to publish uncertainty, failed components, and abstentions. They are not reasons to replace measurement with intuition.

17. Reproducibility package

Before any empirical run, the release package must contain:

- versioned system and environment manifests;
- immutable training-data, code, model, and prompt hashes where disclosure is lawful;
- the boundary and observation contract from Equations 1 and 2;
- component estimators with synthetic positive and negative tests;
- null generators with preserve / destroy assertions;
- architecture-family and world-family split manifests;
- perturbation, sham, rescue, and resource-matching specifications;
- calibration, abstention, and multiplicity rules;
- a machine-readable claim-source matrix;
- a result schema that requires negative and missing results; and
- an independent-review record.

Draft 2 supplied the frozen score-profile smoke test. Draft 3 added `model/estimator-contract-v2.json`, `model/trace_reference_benchmark_v2.py`, `model/trace-reference-benchmark-results-v2.json`, and paired compressed trace samples with a machine-readable schema. Draft 4 preserves those files and adds `model/estimator-contract-v3.json`, `model/trace_reference_benchmark_v3.py`, `model/fixed-adversary-config-v3.json`, deterministic trace samples, and hash-stable validation receipts. These implement trace generation, nine estimators, preserve / destroy nulls, component lesions, an instrumented no-op sham with bounded proxy matches, alternative rescues, development-only threshold calibration, confidence-bound abstention, one validation-configured adversary, and an unseen final-remapping evaluation. They remain synthetic reference implementations, not deployed-system or phenomenal-consciousness validation.

Draft 5 adds a separate learned-family package:

- `model/learned-family-contract-v4.json`;
- `model/architecture-family-manifest-v4.json`;
- `model/world-family-manifest-v4.json`;
- `model/benchmark-custody-v4.json`;
- `model/learned_family_benchmark_v4.py`;
- `model/learned-family-benchmark-results-v4.json`; and
- `model/learned-family-weights-v4.npz`.

The architecture manifest fixes four distinct temporal structures. The world manifest makes every training, unseen-seed, and heldout-remapping row explicit. The custody record says directly that the benchmark is project-local and reproducible but not independently sealed. This resolves an ambiguity in the earlier release plan: independent custody and external implementations remain future validation, while the present preprint reports exactly the synthetic and learned-readout results that were executed.

The evaluator should be runnable by a laboratory that did not build the candidate system. A public benchmark leaderboard must not expose the sealed mechanism labels or worlds used for final evaluation.

18. Conclusion

SAN's 2021 guidebook proposal identified a problem that has become more urgent: artificial systems may imitate signs of consciousness before science can confidently identify consciousness itself. TCNR later supplied a formal relational vocabulary, but its original scalar TCI was too easy to misread and too compensatory to serve as a consciousness meter. The present protocol replaces that shortcut with a typed, adversarial, causal, and abstention-capable battery. Draft 5 preserves the fixed-adversary and instrumented-sham corrections, then tests the frozen battery on four structurally distinct learned-readout temporal systems. Every family fails the full heldout profile while retaining only the relational template component. That

negative result strengthens the central claim: relation-level evidence must remain typed, noncompensatory, and subordinate to heldout causal and functional tests.

Its strongest possible conclusion is precise: under a declared boundary, a system implements a measurable SAN / TCNR mechanism profile that generalizes across held-out families and responds to selective perturbation and rescue as predicted. That would be significant evidence about mechanism. It would still not be a proof of subjective experience. Keeping those claims separate is what makes the guidebook scientifically useful and ethically serious.

Authorship and research history

The originating SAN concepts, machine-consciousness guidebook proposal, artificial-neurology program, multimodal coincidence account, self-as-direction relation, and associated ethical concern were developed and recorded by Micah Blumberg before ChatGPT became publicly available. Later AI assistance was used under Micah Blumberg's direction for editorial development, mathematical formalization, literature routing, source checking, and software validation. AI dialogue is not an independent scientific source and is not represented as the origin of the theory.

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